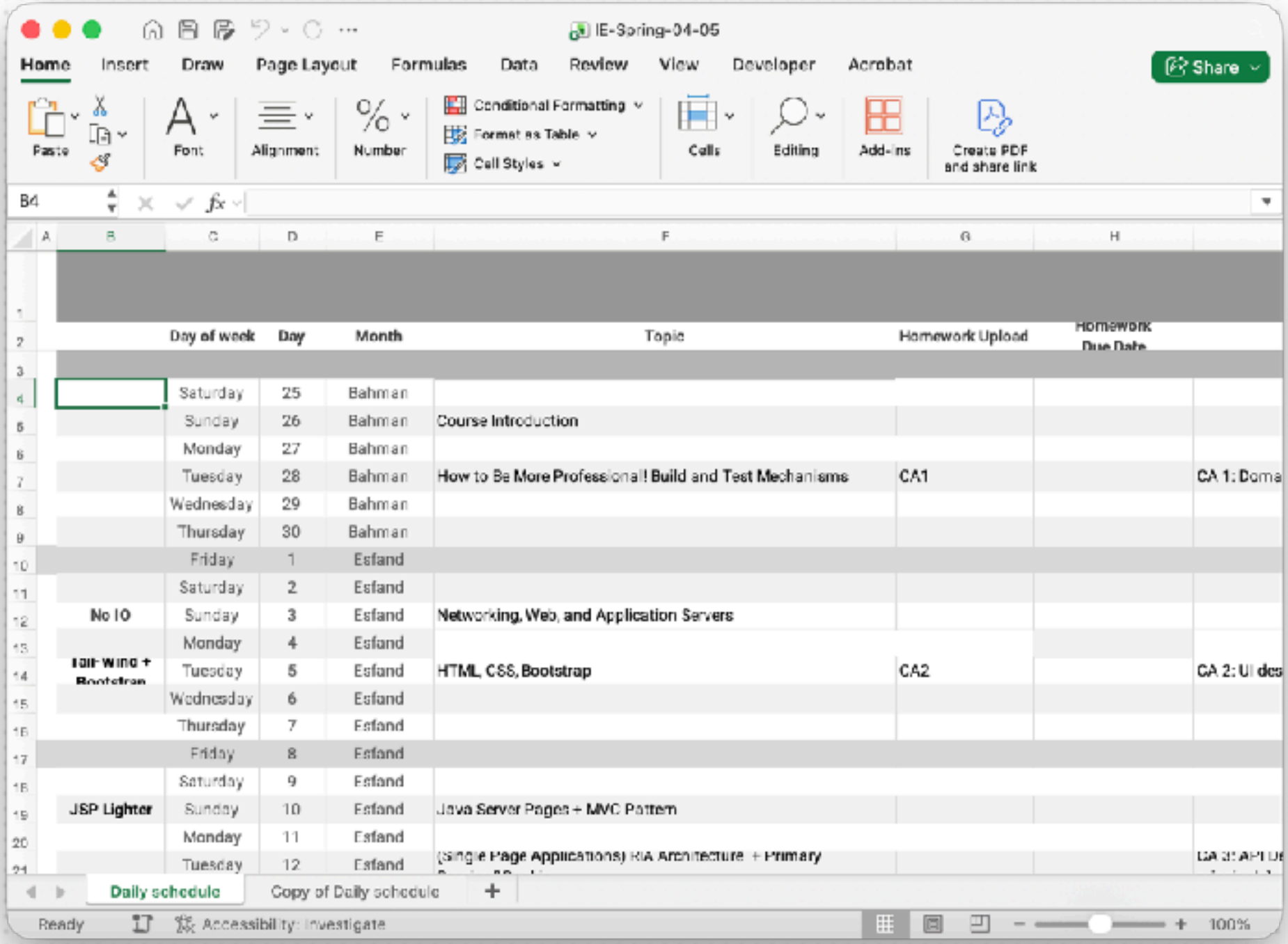
Introduction to Internet Engineering

Ehsan Khamespanah

Spring 2026

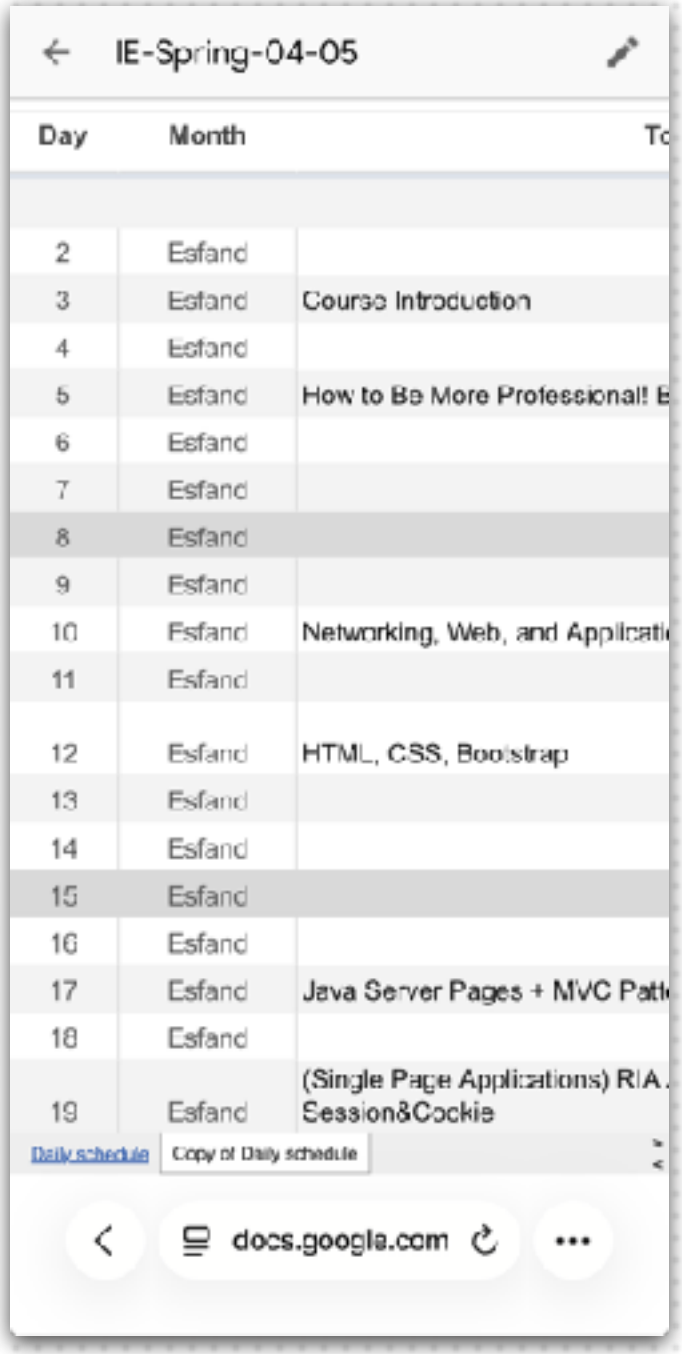

FDISK Options
Current fixed disk drive: 1
Choose one of the following:
1. Create DOS partition or Logical DOS Drive
2. Set active partition
3. Delete partition or Logical DOS Drive
4. Display partition information
Enter choice: [1]
Press Esc to exit FDISK


IE-Spring-04-05
Home
Insert
Draw
Page Layout
Formulas
Data
Review
View
Developer
Acrobat
Share
Paste
Font
Alignment
Number
Conditional Formatting
Format as Table
Cell Styles
Cells
Editing
Add-ins
Create PDF and share link
B4
Day of week
Day
Month
Topic
Homework Upload
Homework Due Date
Saturday 25 Bahman
Sunday 26 Bahman Course Introduction
Monday 27 Bahman
Tuesday 28 Bahman How to Be More Professional! Build and Test Mechanisms CA1 CA 1: Dome
Wednesday 29 Bahman
Thursday 30 Bahman
Friday 1 Esfand
Saturday 2 Esfand
No IO Sunday 3 Esfand Networking, Web, and Application Servers
Monday 4 Esfand
Tuesday 5 Esfand HTML, CSS, Bootstrap CA2 CA 2: UI des
Wednesday 6 Esfand
Thursday 7 Esfand
Friday 8 Esfand
Saturday 9 Esfand
JSP Lighter Sunday 10 Esfand Java Server Pages + MVC Pattern
Monday 11 Esfand
Tuesday 12 Esfand
Daily schedule
Copy of Daily schedule
Ready
Accessibility: Investigate
100%

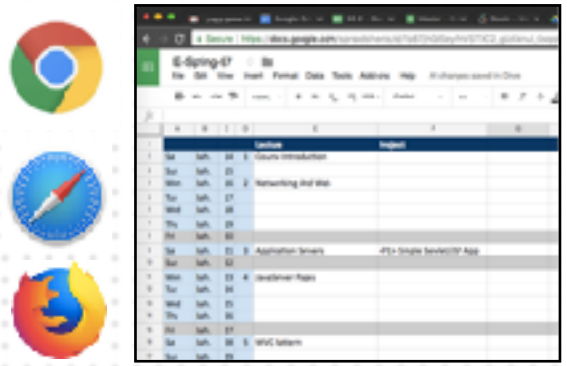
IE-Spring-04-05

File Edit View Insert Format Data Tools Extensions Help

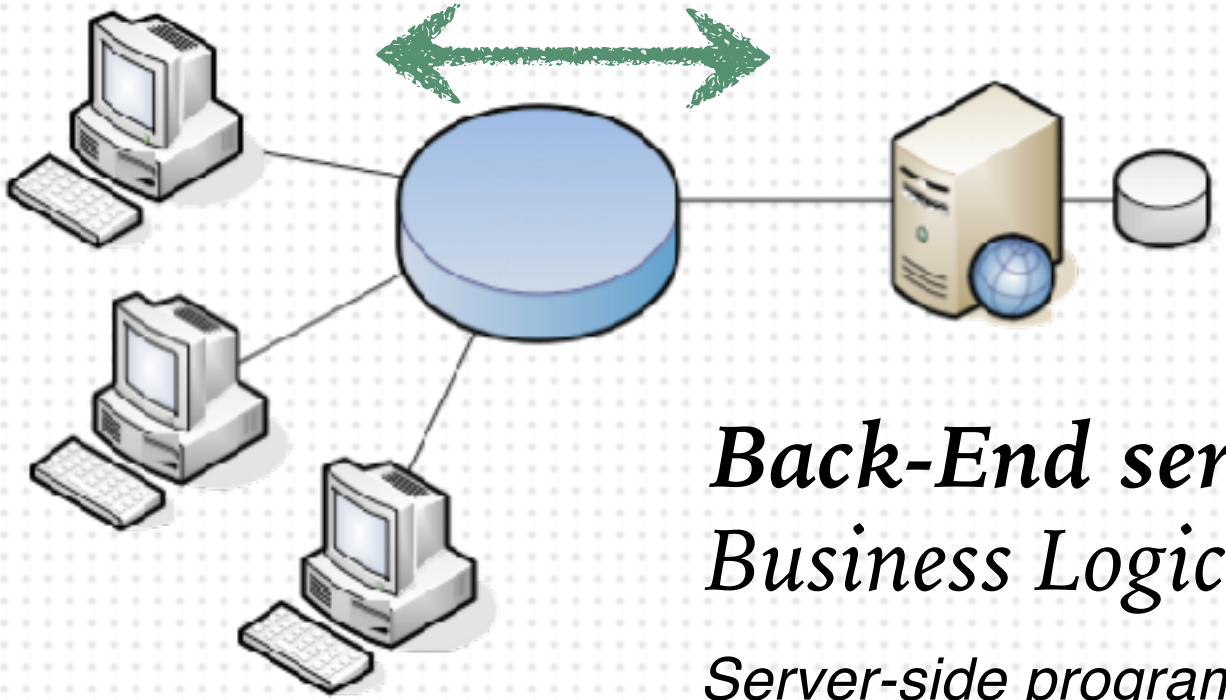
	Day of week	Day	Month	Topic	Homework Upload
	Saturday	2	Esfand		
	Sunday	3	Esfand	Course Introduction	
	Monday	4	Esfand		
	Tuesday	5	Esfand	How to Be More Professional! Build and Test Mechanisms	
	Wednesday	6	Esfand		
	Thursday	7	Esfand		
	Friday	8	Esfand		
	Saturday	9	Esfand		
No IO	Sunday	10	Esfand	Networking, Web, and Application Servers	
	Monday	11	Esfand		
Tail-Wind +	Tuesday	12	Esfand	HTML, CSS, Bootstrap	
	Wednesday	13	Esfand		
	Thursday	14	Esfand		
	Friday	15	Esfand		
	Saturday	16	Esfand		
JSP Lighter	Sunday	17	Esfand	Java Server Pages + MVC Pattern	
	Monday	18	Esfand		
	Tuesday	19	Esfand	(Single Page Applications) RIA Architecture + Primary	
	Wednesday	20	Esfand		



Different Ecosystem



Web API
(SOAP/REST/POX/...)





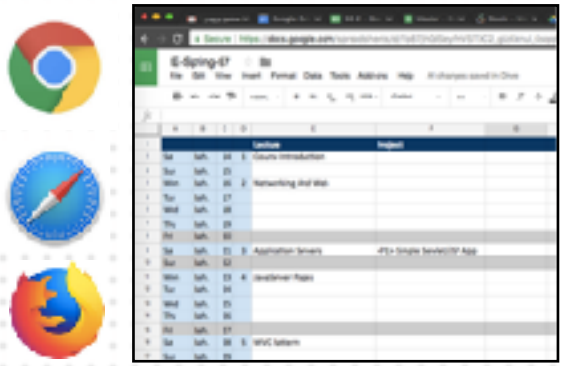
Client Applications
Web Pages or Mobile Apps

Back-End servers
Business Logic Services

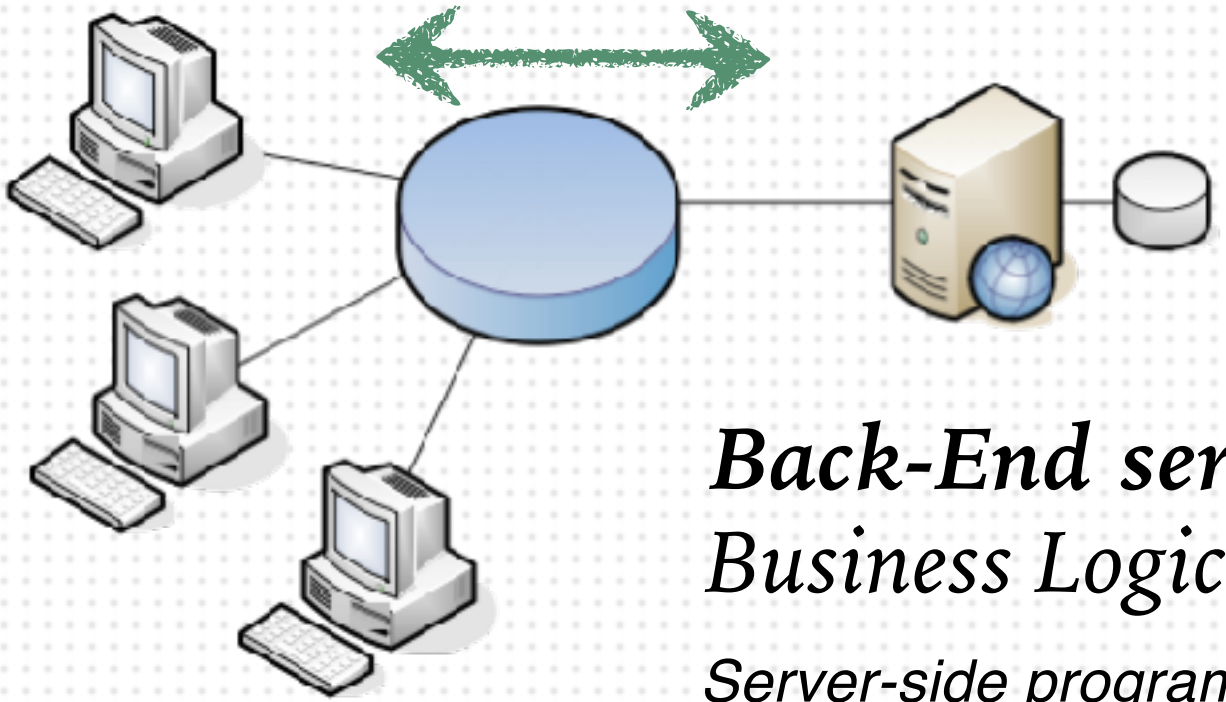
Server-side programs providing business services over the web

Different Ecosystem

Is that all?



Web API
(SOAP/REST/POX/...)



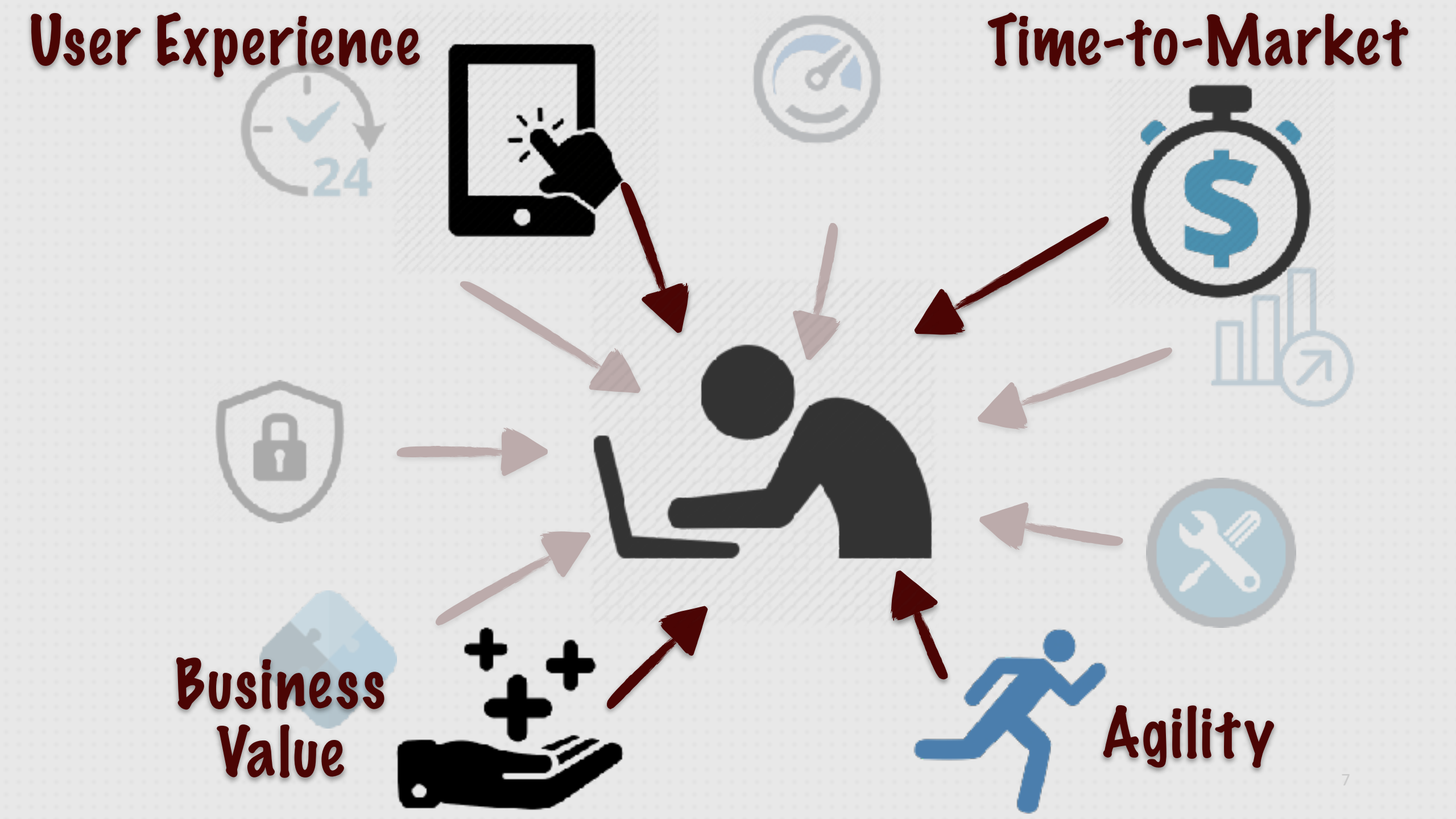


Client Applications
Web Pages or Mobile Apps

Back-End servers
Business Logic Services

Server-side programs providing business services over the web


Availability
Performance
Scalability
Security
Interoperability
Maintainability


User Experience
Time-to-Market
Business Value
Agility

Goals of This Course

- To learn how to program for the Internet / Web
  - To learn the Internet-scale application architectural issues and patterns
- Although you will need more experience to be an architect of such applications
- To become familiar with new trends in web/enterprise application development

Main Topics

- Web basics (Networking, HTTP, Containers, HTML/CSS)
- Java web programming (Servlets, JSPs, Web APIs)
- Rich Internet Applications (JavaScript, AJAX, React)
- Enterprise Application Architecture
  (Layering, Data Access, Concurrency, …)
- Infrastructure (Build and Release, CI/CD)
- Security (Security Threats, OWASP)
- Many other topics!

Languages, Frameworks, Tools













Apache Tomcat





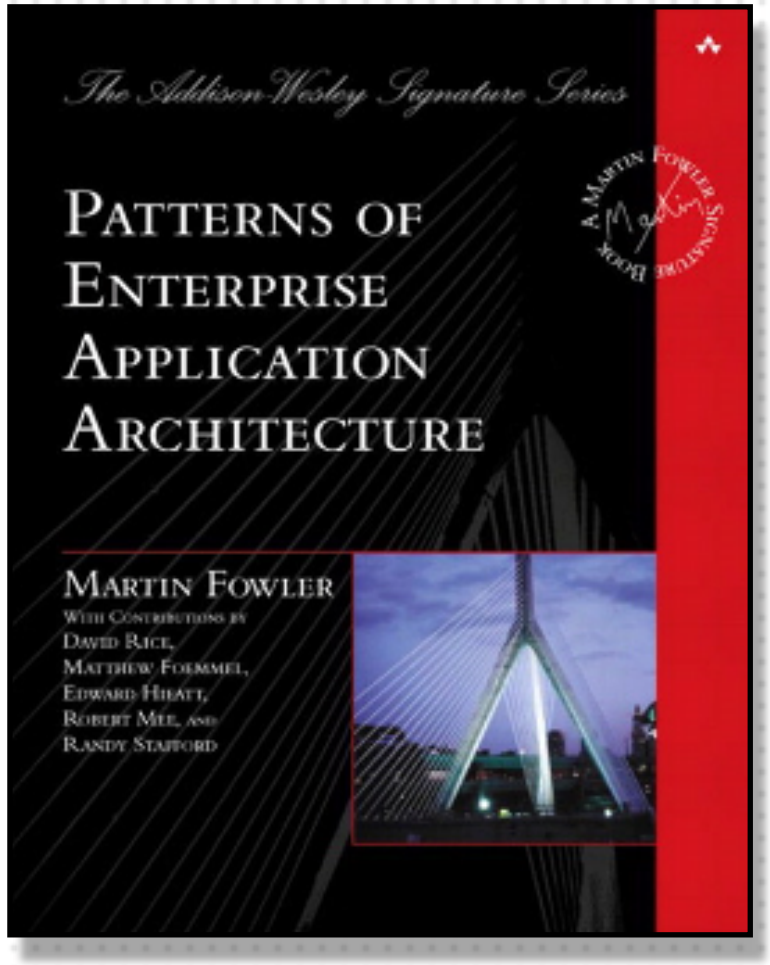
HIBERNATE

Prerequisites

- Advanced Programming
- Database Basics: the relational model + SQL
- Computer Network Basics

Resources

- Technology parts:
  - Online resources
- Architectural parts:
  - Fowler, Patterns of Enterprise Application Architecture
  - Online resources



Grading Policy

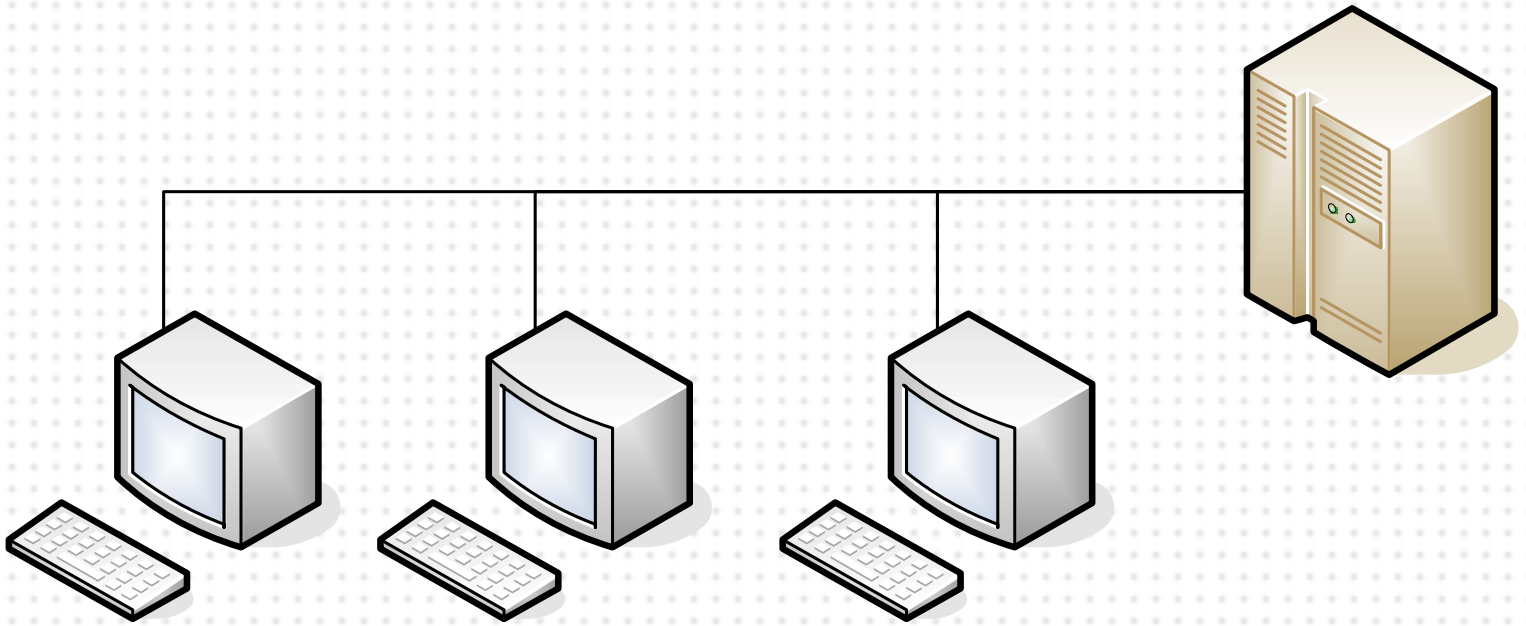
Item	Weight
Midterm Exam	4 pts.
Final Exam	5 pts.
Computer Assignments	12 pts.

- At least 4.5 points from the exams
- Assignments are to be performed in teams of two
- Late Policy: 4 days for each team
- Copied assignments: both sides loose 4 pts.

Mainframes Era



Mainframes Era



mainframe

handles all the processing

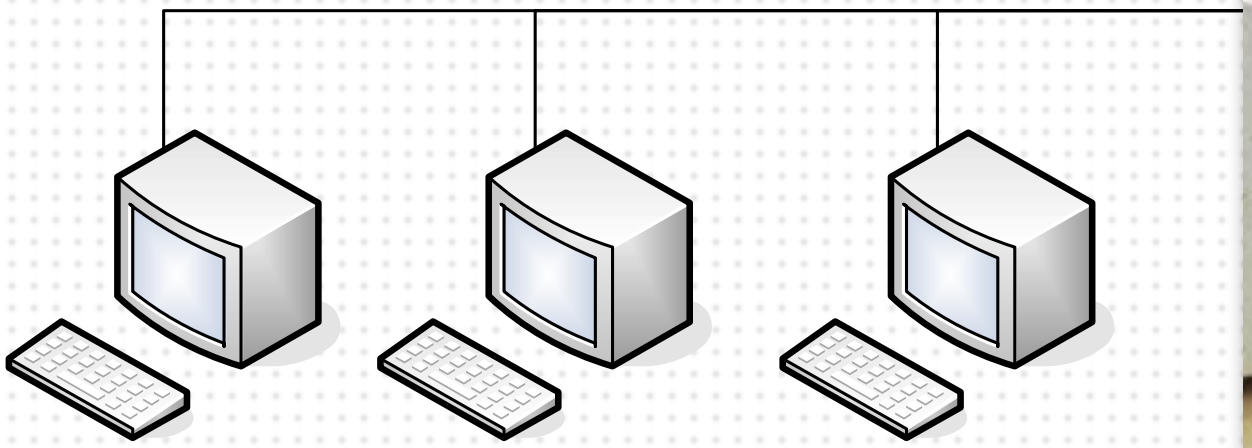
Capable of handling and processing very large amounts of data quickly

terminals

only handle input and output

Very limited processing (e.g. line editing)

Mainframes Era



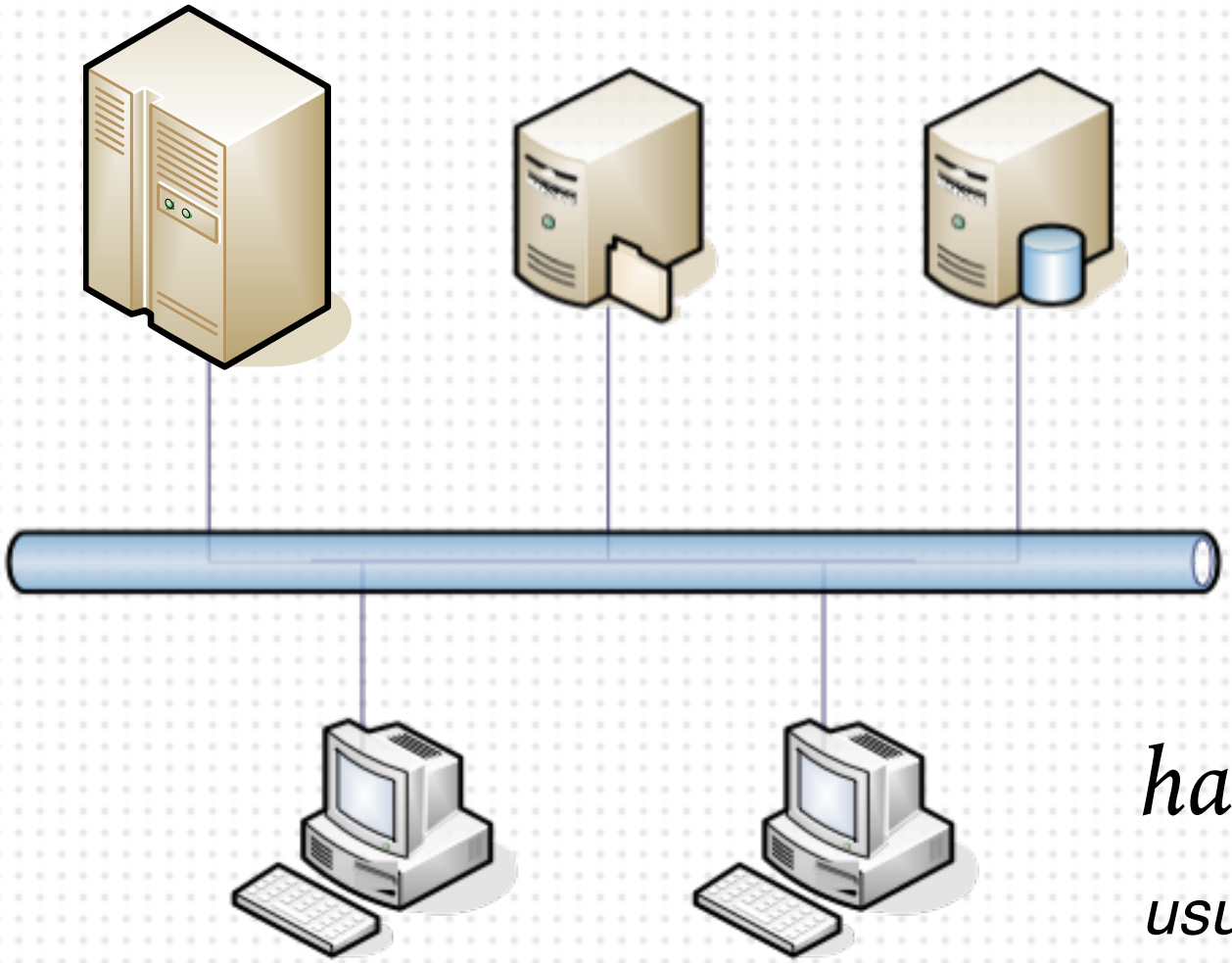


terminals

only handle input and output

Very limited processing (e.g. line editing)

Clever Clients



servers

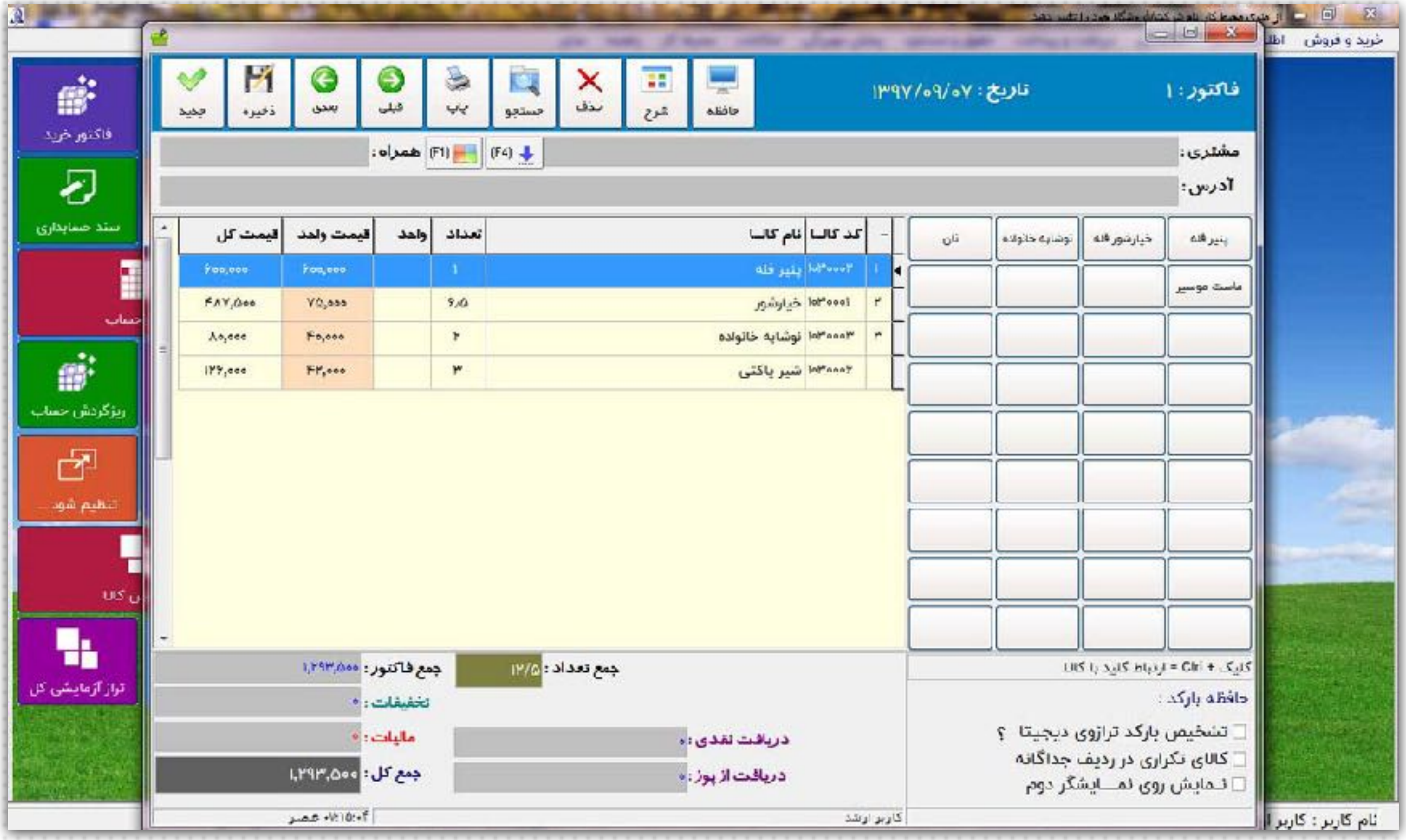
provide various types of services (data, file, etc.)

mainframe and mid-range computers

clients

handle part of the processing

usually desktop PCs -- way more powerful than terminals

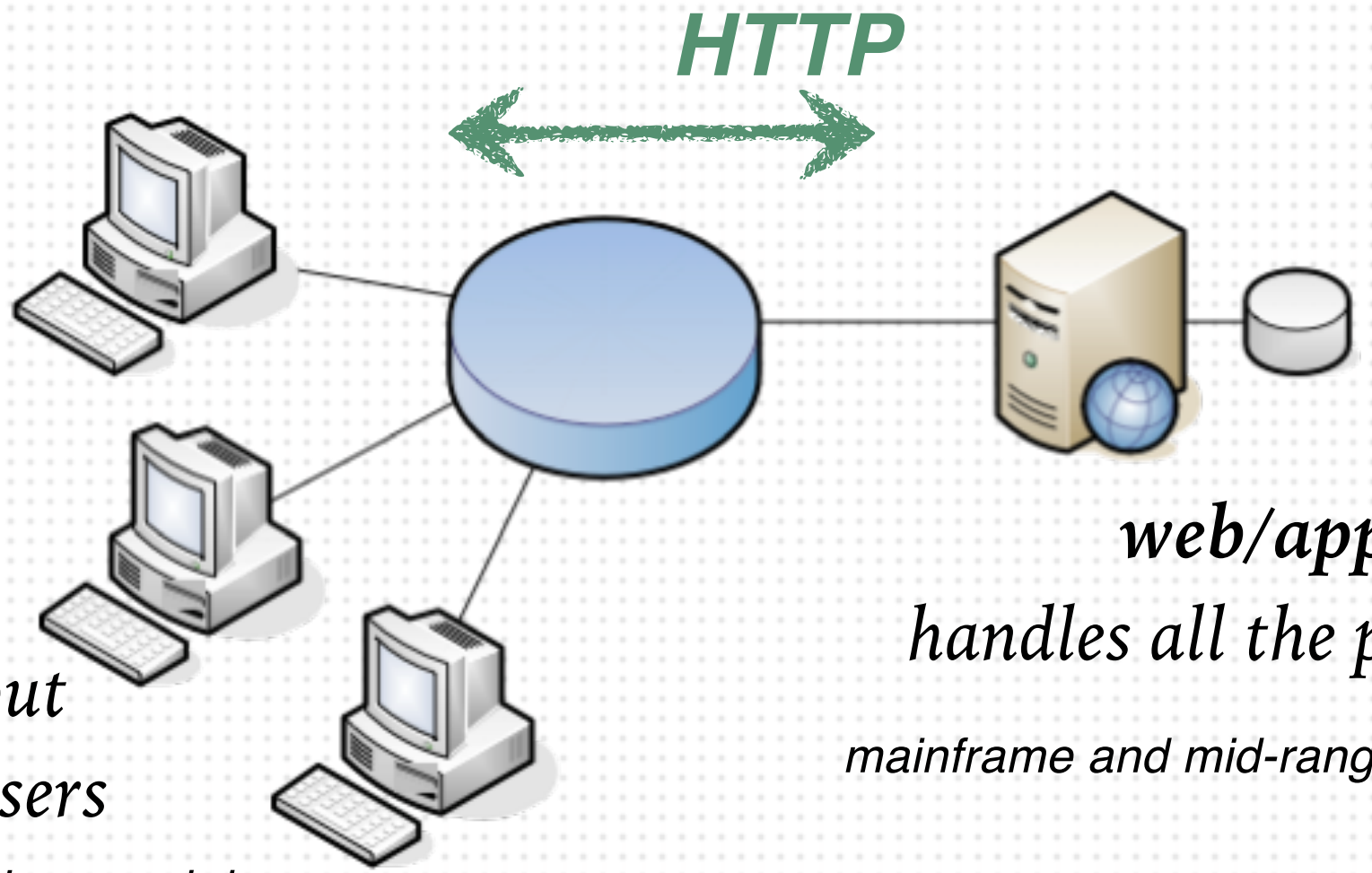

فاکتور خرید
سند حسابداری
حساب
ویرگردش حساب
تنظیم شود
تراز آزمایشی کل
خرید و فروش
فاکتور : ۱
تاریخ : ۱۳۹۷/۰۹/۰۷
حافظه
شرح
حذف
جستجو
چاپ
قبلی
بعدی
ذخیره
جدید
مشتری :
آدرس :
همراه :
(F1)
(F4)
کد کالا
نام کالا
تعداد
واحد
قیمت واحد
قیمت کل
۱ ۱۵۳۰۰۰۲ پنیر فله ۱ ۶۰۰,۰۰۰ ۶۰۰,۰۰۰
۲ ۱۵۳۰۰۰۱ خیارشور ۶/۵ ۷۵,۰۰۰ ۴۸۷,۵۰۰
۳ ۱۵۳۰۰۰۳ نوشابه خانواده ۲ ۴۰,۰۰۰ ۸۰,۰۰۰
۱۵۳۰۰۰۲ شیر پاکتی ۳ ۴۲,۰۰۰ ۱۲۶,۰۰۰
پنیر فله
خیارشور فله
نوشابه خانواده
نان
ماست موسیر
جمع تعداد : ۱۲/۵
جمع فاکتور : ۱,۲۹۳,۵۰۰
تخفیفات : ۰
مالیات : ۰
دریافت نقدی : ۰
دریافت از پوز : ۰
جمع کل : ۱,۲۹۳,۵۰۰
حافظه بارکد :
تشخیص بارکد ترازوی دیجیتا ؟
کالای تکراری در ردیف جداگانه
نمایش روی نمایشگر دوم
نام کاربر : کاربر ا

Client-Server Architecture

- Client applications with graphical UI
- Custom client-server protocols
- Various network architectures
- Platform-specific client applications
- High administration cost of the clients

Web Applications





clients

HTML input/output through web browsers

personal computers running a web browser

web/app servers

handles all the processing

mainframe and mid-range computers

Benefits of Web Applications

- Extensive use of standards: HTML, HTTP, ...
- You only need an HTML-capable browser, everything else is done by the server
- Virtually zero client administration cost
- Instantaneous application updates



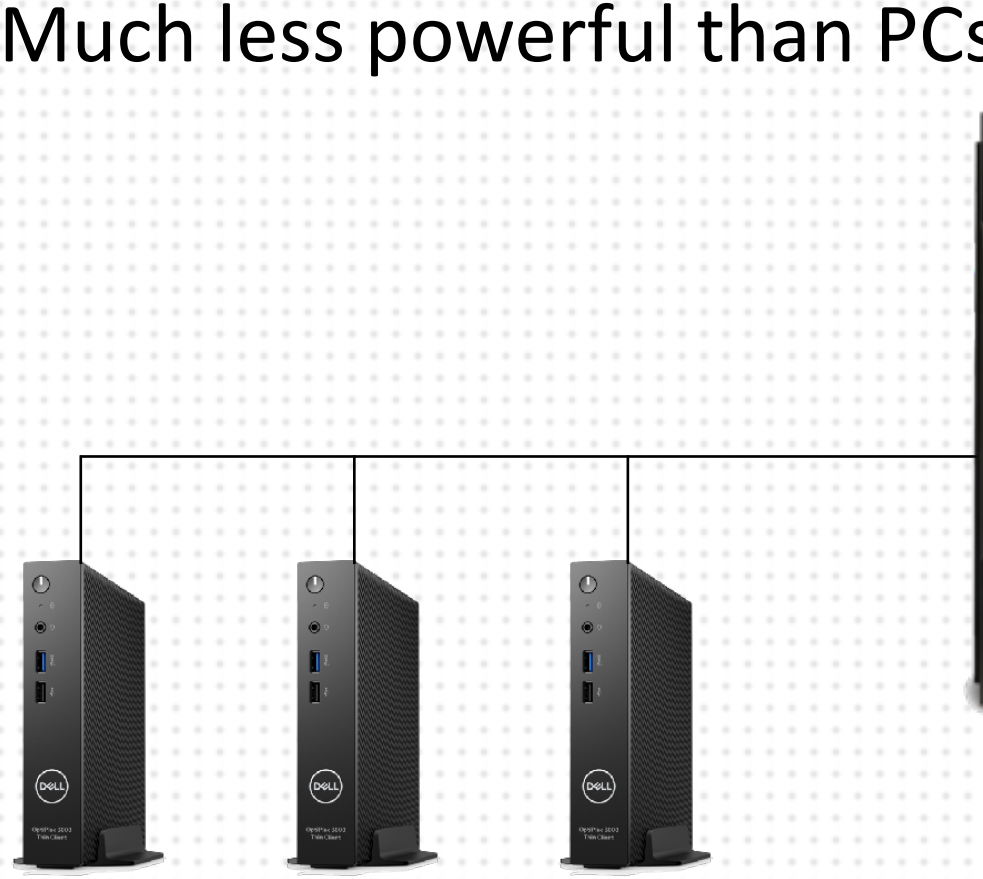
Thin Clients

- The idea: you have the browser and you don’t need more.
- Much less powerful than PCs



Thin Clients

- The idea: you have the browser and you don’t need more.
- Much less powerful than PCs







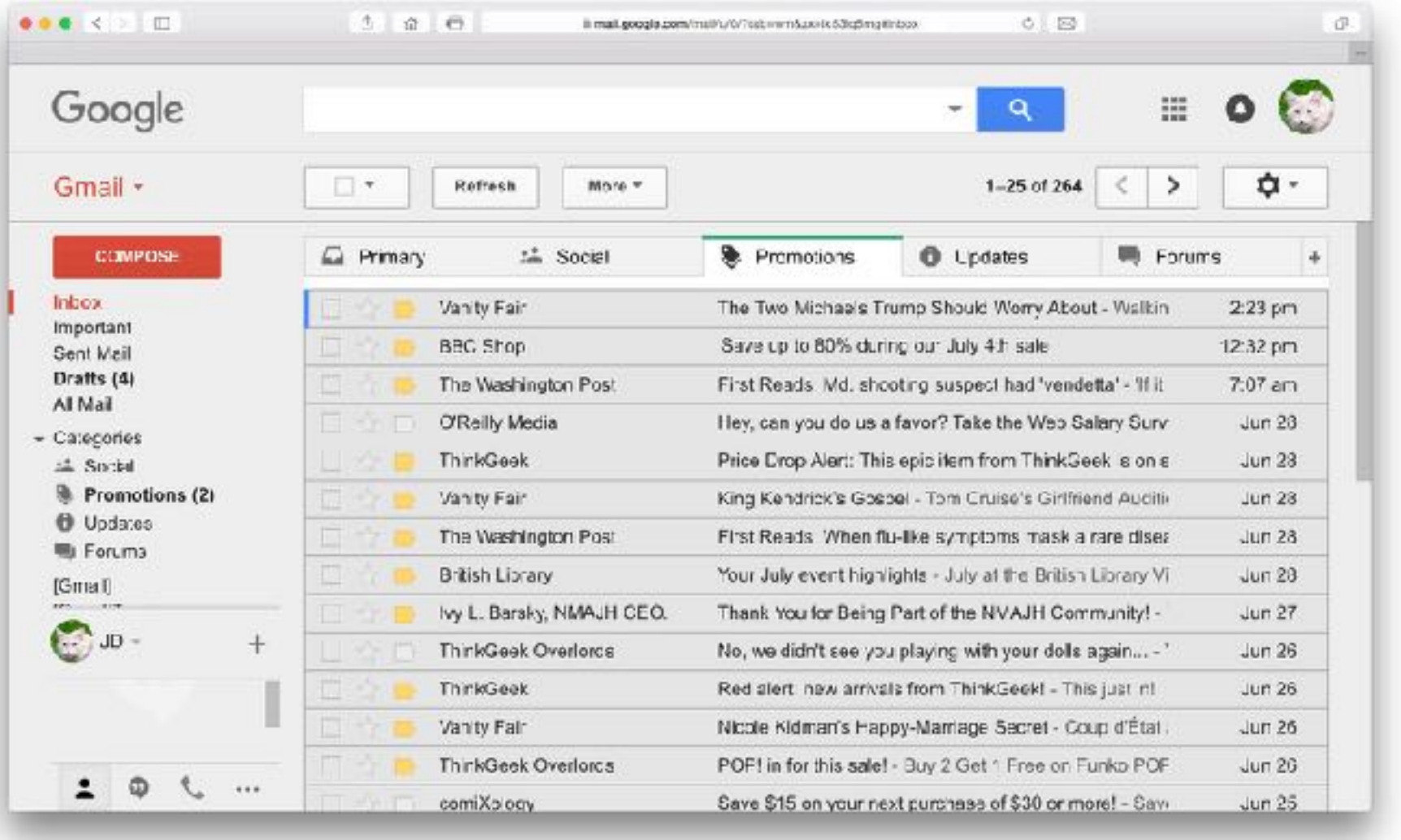
Thin Clients

- The idea: you have the browser and you don’t need more.
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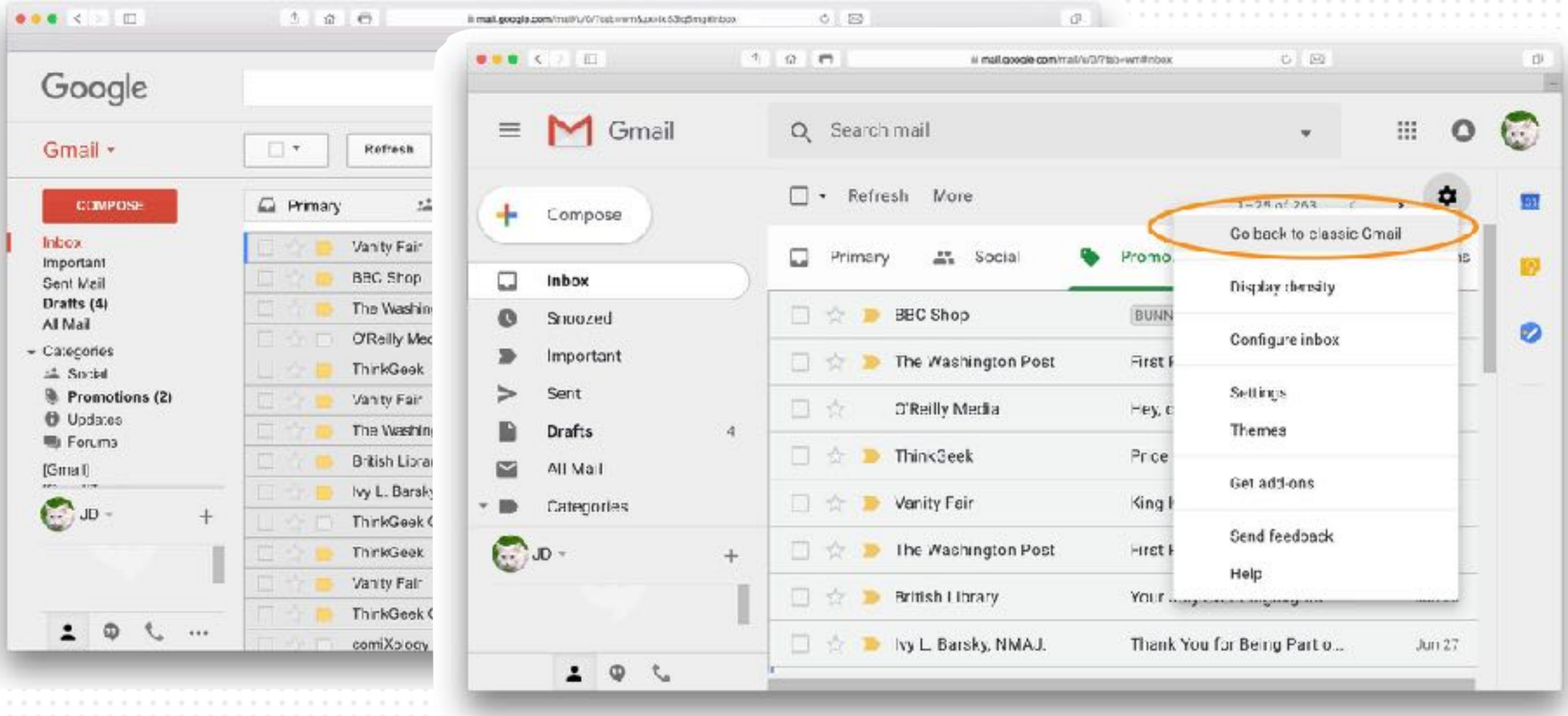




Improving the Interaction



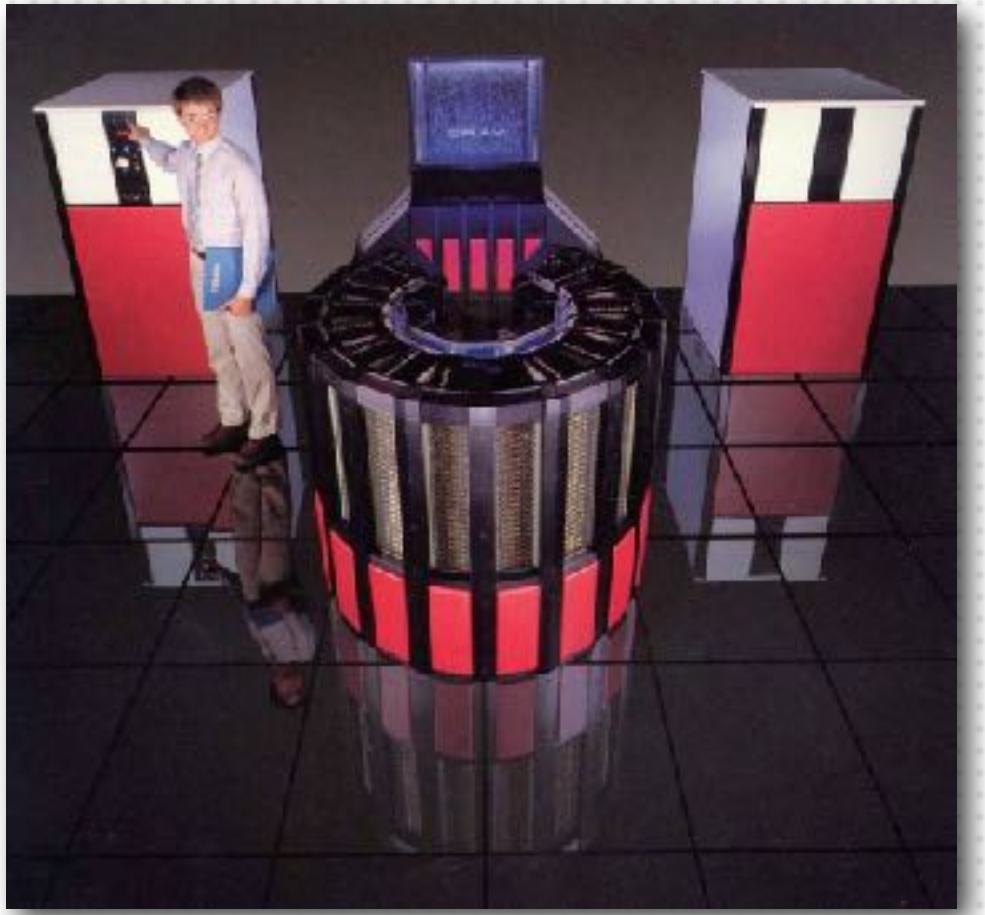
Improving the Interaction



Enabling Technologies

- JavaScript (+ AJAX, jQuery, JSON, etc.)
- Client-side components (Flash, ActiveX, Applets, etc.)
- HTML 5 + CSS





1986 IBM Cray-2 Supercomputer

2011 Apple iPad-2 Tablet



2017 Apple Watch

